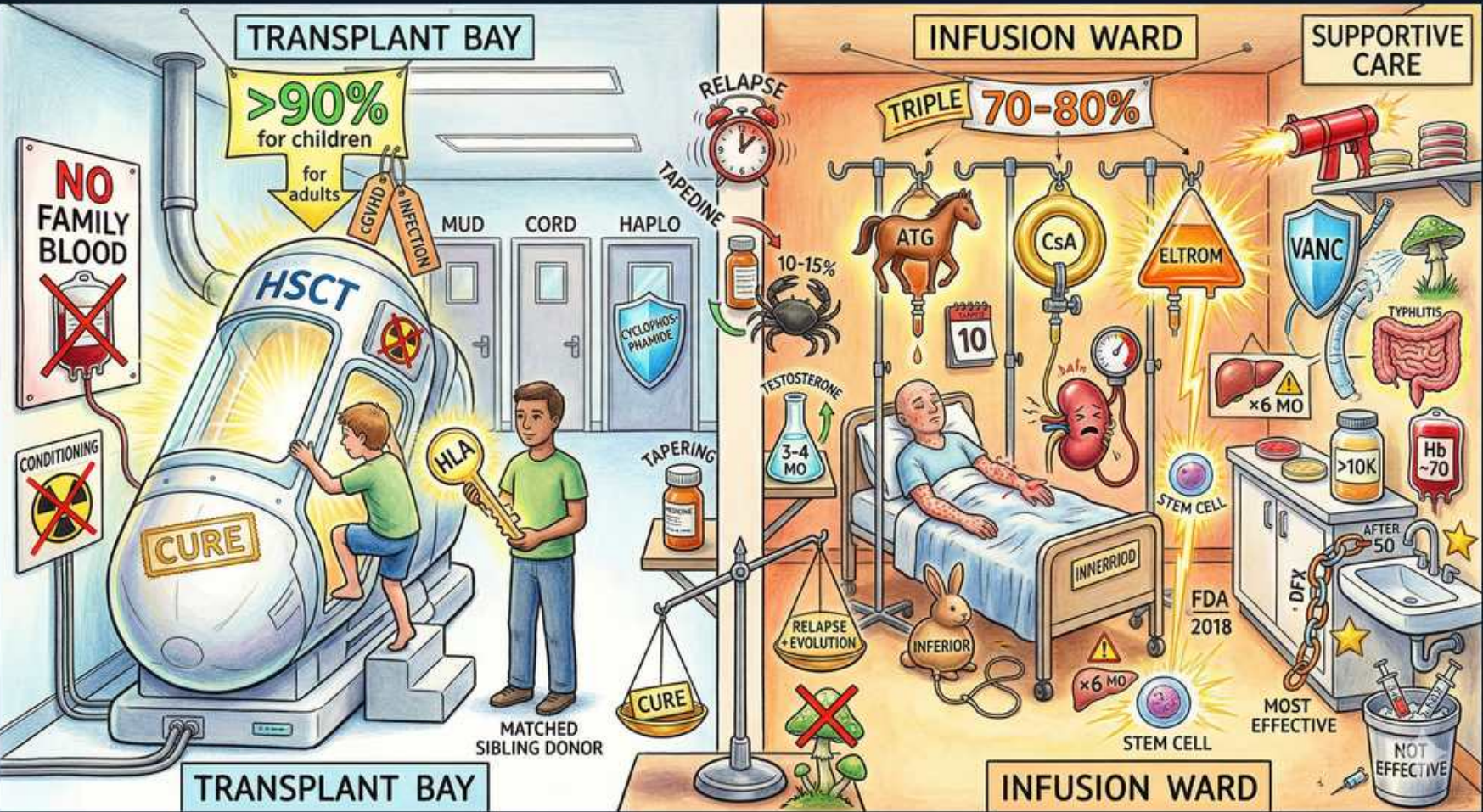


Bone Marrow Failure — The Rescue Station (AA Treatment)





## 1. HSCT transplant bay

### WHAT YOU SEE

TRANSPLANT BAY with an HSCT rocket-pod and a child

### WHAT IT ENCODES

Allogeneic HEMATOPOIETIC STEM CELL TRANSPLANT is FIRST-LINE for YOUNG patients (roughly <40–50 yr, fit) with SEVERE or VERY-SEVERE aplastic anemia who have a matched sibling donor. It is the only potentially CURATIVE therapy because it replaces the failed marrow.

### BOARD TRAP

Transplant is NOT first-line for everyone — age and donor availability decide. Older or unfit patients, or those without a matched donor, get immunosuppressive therapy (IST) first. Choosing 'HSCT for all' is the classic distractor.

## 2. >90% children cure

### WHAT YOU SEE

Green banner >90% for children

### WHAT IT ENCODES

Matched-sibling allo-HSCT cures >90% of children and young adults with severe AA, with the best long-term outcomes and (unlike IST) low risk of later clonal evolution to MDS/AML.

### BOARD TRAP

Younger age = better transplant outcome and lower GVHD risk; this is precisely why upfront transplant is favored in the young while IST is preferred in older adults.

## 3. Matched sibling donor

### WHAT YOU SEE

MATCHED SIBLING DONOR figure handing over an HLA key labeled CURE

### WHAT IT ENCODES

The preferred graft source is an HLA-MATCHED SIBLING DONOR — best engraftment, least GVHD. Its availability (plus patient age/severity) is the pivotal branch point in the treatment algorithm.

### BOARD TRAP

Young + severe AA + matched sibling → transplant first. No matched sibling → start IST while searching for an unrelated/alternative donor (matched-unrelated transplant is increasingly used upfront in the young when a quick MUD is available).

## 4. Donor doors MUD/CORD/HAPLO

### WHAT YOU SEE

Doors labeled MUD, CORD, HAPLO

### WHAT IT ENCODES

When no matched sibling exists, alternative donor sources are matched UNRELATED donor (MUD), CORD blood, or HAPLOIDENTICAL donor — used for transplant after IST failure, or upfront in young patients in experienced centers.

### BOARD TRAP

Alternative-donor transplants carry higher GVHD/graft-failure risk than matched-sibling grafts, so they are generally reserved for IST-refractory young patients rather than offered before a trial of immunosuppression.

## 5. No family blood

### WHAT YOU SEE

Red NO FAMILY BLOOD sign with crossed bag

### WHAT IT ENCODES

Avoid transfusing blood products from FAMILY MEMBERS in any AA patient who might be transplanted — shared HLA antigens increase alloimmunization and transfusion-associated GVHD and can compromise later engraftment.

### BOARD TRAP

Counterintuitive but high-yield: family donors are WORSE, not better, for transfusion in a potential transplant candidate. Also minimize transfusions overall pre-transplant.

## 6. Conditioning radiation

### WHAT YOU SEE

CONDITIONING radiation symbol on the pod

### WHAT IT ENCODES

A CONDITIONING regimen (e.g., cyclophosphamide ± ATG; radiation in some protocols) precedes the graft to ablate the recipient's autoreactive immune system and allow engraftment.

### BOARD TRAP

Minimize pretransplant transfusions and use leukoreduced/irradiated products to reduce alloimmunization and graft rejection — sensitization from prior transfusions worsens engraftment.

## 7. ATG + Cyclosporine

### WHAT YOU SEE

INFUSION WARD IV poles labeled ATG and CsA

### WHAT IT ENCODES

IMMUNOSUPPRESSIVE THERAPY (IST) = antithymocyte globulin (ATG, horse/equine ATG preferred over rabbit) PLUS cyclosporine. It is FIRST-LINE for patients who are NOT transplant candidates: older adults, no matched sibling donor, or significant comorbidities. It suppresses the T-cell attack so endogenous marrow recovers.

### BOARD TRAP

IST is the answer for OLDER/unfit patients or young patients lacking a donor — NOT first-line for a young, fit patient WITH a matched sibling (that patient gets transplant). Horse ATG outperforms rabbit ATG in AA.



## 8. Triple 70-80%

### WHAT YOU SEE

TRIPLE 70-80% banner over the infusion ward

### WHAT IT ENCODES

TRIPLE therapy (ATG + cyclosporine + eltrombopag) achieves response rates of ~70–80%, higher and faster than ATG+cyclosporine alone. Responses take weeks to months, so supportive care must continue meanwhile.

### BOARD TRAP

IST improves counts but is NOT curative and does not replace the marrow — relapse and clonal evolution remain possible, requiring long-term follow-up.

## 9. Eltrombopag

### WHAT YOU SEE

ELTROM IV bag, FDA 2018 tag

### WHAT IT ENCODES

ELTROMBOPAG is an oral THROMBOPOIETIN-RECEPTOR AGONIST that stimulates surviving stem cells; added to ATG+cyclosporine it boosts overall and complete response rates and speeds count recovery (approved upfront with IST in 2018).

### BOARD TRAP

Mechanism is TPO-receptor agonism on the residual stem-cell pool — not immunosuppression. Monitor for clonal cytogenetic changes/evolution while on therapy.

## 10. Relapse / clonal evolution

### WHAT YOU SEE

Clock RELAPSE 10-15% and RELAPSE/EVOLUTION scale

### WHAT IT ENCODES

After IST, ~10–15%+ relapse and there is a real long-term risk of CLONAL EVOLUTION to MDS, AML, or clinically significant PNH. Lifelong monitoring (CBC, flow for PNH, marrow/cytogenetics) is required.

### BOARD TRAP

IST is NOT permanently curative — its major drawback versus transplant is relapse and late clonal evolution. This is a key reason upfront transplant is favored in the young.

## 11. Supportive transfusion

### WHAT YOU SEE

SUPPORTIVE CARE area with transfusion and Hb >7 target

### WHAT IT ENCODES

SUPPORTIVE CARE underpins all management: RBC transfusion for symptomatic anemia (restrictive threshold ~Hb 7 g/dL) and platelet transfusion for bleeding/very low counts — all products LEUKOREduced and IRRADIATED to limit alloimmunization and transfusion-associated GVHD.

### BOARD TRAP

Always use leukoreduced + irradiated products and avoid family donors in transplant candidates; over-transfusion before transplant promotes alloimmunization and graft failure.

## 12. Antimicrobials VANC

### WHAT YOU SEE

VANC bottle and antimicrobial icons, TYPHLITIS

### WHAT IT ENCODES

Neutropenic patients need aggressive infection management: empiric BROAD-SPECTRUM antibiotics for febrile neutropenia, antifungal therapy for persistent fever, and vigilance for neutropenic enterocolitis (TYPHLITIS). Prophylaxis may be used in prolonged severe neutropenia.

### BOARD TRAP

Febrile neutropenia is an emergency — empiric anti-pseudomonal antibiotics immediately. Avoid routine prophylactic platelet/granulocyte myths; G-CSF gives only transient, limited benefit and does not treat the underlying disease.

## 13. Iron chelation DFX

### WHAT YOU SEE

DFX chelator icon near transfusion bucket

### WHAT IT ENCODES

Chronic RBC transfusion causes IRON OVERLOAD; treat with iron chelation (DEFERASIROX/DFX, oral; or deferoxamine) once ferritin/transfusion burden is high, to prevent cardiac/hepatic/endocrine iron toxicity.

### BOARD TRAP

Iron overload is from transfusions, not the disease itself — monitor ferritin and chelate transfusion-dependent patients. Relevant especially in IST-treated/relapsed patients with ongoing transfusion need.

## 14. Most effective = transplant

### WHAT YOU SEE

CURE key on scale vs INFERIOR rabbit, MOST EFFECTIVE label

### WHAT IT ENCODES

Bottom line: allo-HSCT is the ONLY CURATIVE option and the most effective definitive therapy (the 'CURE' key), while IST is non-curative but is preferred when transplant is too risky (older/unfit patients or no donor). The decision balances age, disease severity, and donor availability.

### BOARD TRAP

Both can be 'right' depending on the patient: transplant for young + severe + matched sibling; IST for older/unfit or no donor. The exam tests this branching, not a single universal first-line.